

Preface

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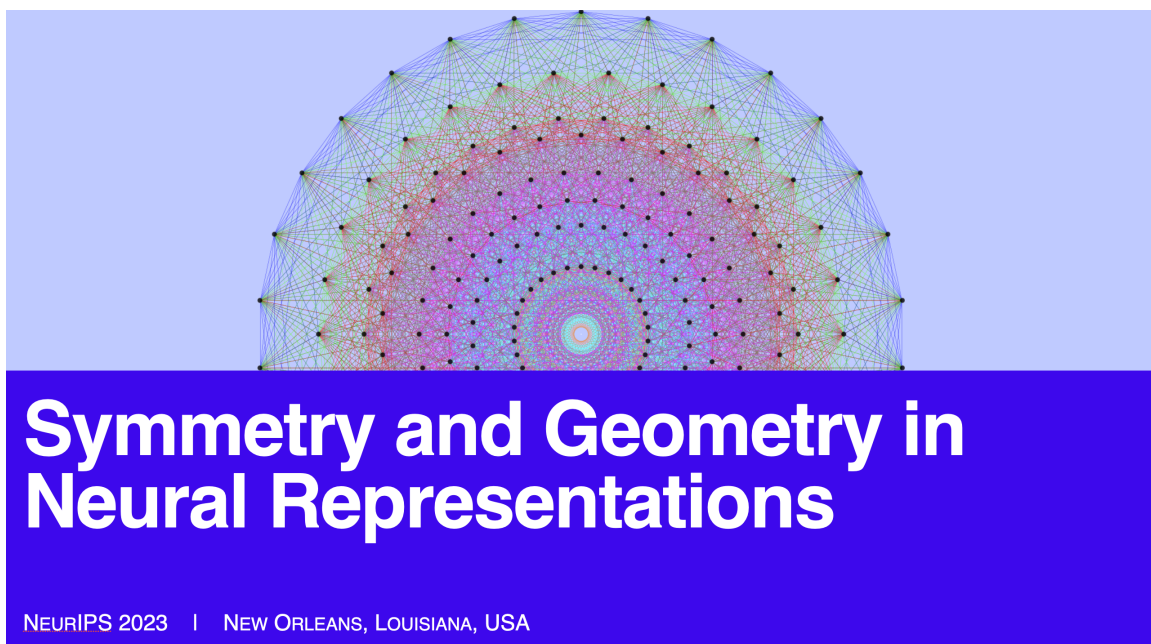
Editors: Francisco Acosta, Simone Azeglio, Chase van de Geijn, Christian Shewmake, Sophia Sanborn, Nina Miolane

1. Symmetry and Geometry in Neural Representations

An emerging set of findings in sensory and motor neuroscience is beginning to illuminate a new paradigm for understanding the neural code. Across sensory and motor regions of the brain, neural circuits are found to mirror the geometric and topological structure of the systems they represent—either in their synaptic structure, or in the implicit manifold generated by their activity. This phenomenon can be observed in the circuit of neurons representing head direction in the fly [Kim et al. \(2017\)](#); [Wolff et al. \(2015\)](#), in the activities of grid cells [Chaudhuri et al. \(2019\)](#); [Gardner et al. \(2022\)](#), and in the low-dimensional manifold structure observed in motor cortex [Gallego et al. \(2017\)](#). This suggests a general computational strategy that is employed throughout the brain to preserve the geometric structure of data throughout stages of information processing.

Independently but convergently, this very same computational strategy has emerged in the field of deep learning. The nascent sub-field of Geometric Deep Learning [Bronstein et al. \(2021\)](#) incorporates geometric priors into artificial neural networks to preserve the geometry of signals as they are passed through layers of the network. This approach provably demonstrates gains in the computational efficiency, robustness, and generalization performance of these models. More recently, principles such as sparsity, equivariance, and compositionality are guiding the development of more generalizable and interpretable models, including new approaches to foundation model distillation and to the mechanistic interpretability of large-scale networks.

The convergence of these findings suggests deep, substrate-agnostic principles for information processing. Symmetry and geometry were instrumental in unifying models of fundamental forces and elementary particles in 20th-century physics. Likewise, they have the potential to illuminate unifying principles for how neural systems form useful representations of the world.



2. The Workshop

The fourth annual [NeurIPS Workshop on Symmetry and Geometry in Neural Representations \(NeurReps\)](#) was conceived to bring together researchers at the nexus of applied geometry, deep learning, and neuroscience, with the goal of advancing this understanding and illuminating geometric principles for neural information processing. This year’s workshop expanded into emerging frontiers, including geometric mechanistic interpretability, the geometry of representations in foundation models of brain activity, and improvements to LLM design guided by a geometric understanding of transformers. Ultimately, we hope that this venue and associated community will support the development of the geometric approach to understanding neural representations, while strengthening ties to the mathematics community. The [Neural Information Processing Systems \(NeurIPS\)](#) conference historically emerged from the field of theoretical neuroscience or “connectionism.” This venue thus further serves the workshop’s goal of reinforcing the bond between deep learning and neuroscience.

2.1. Call for Papers

Our call for papers invited submissions for publication in this volume and presentation at the conference. The call requested submissions contributing novel research at the intersection of geometric deep learning, computational neuroscience, geometric statistics, and topological data analysis, which incorporate symmetry, geometry, or topology into the *design of artificial neural networks*, the *analysis of neural data*, or *theories of neural computation*. Both theoretical contributions and applied results were encouraged, as well as the use of diverse mathematical objects such as quotient spaces, fiber bundles, Lie groups, Riemannian manifolds, graphs, and group representations. Two tracks were established:

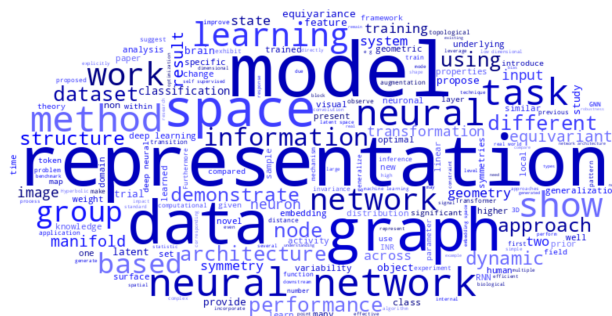


Figure 1: A word cloud constructed from the abstracts of all submissions.

1. **Proceedings Track:** NeurReps Proceedings papers are up to 10 pages long, excluding references and appendices. This track is intended for self-contained research papers with a high degree of development. Accepted papers are published in this volume.
2. **Extended Abstract Track:** Extended abstracts are up to 5 pages long, excluding references and appendices. This track is a space for contributions such as early-stage results, insightful negative findings, opinion pieces, or novel datasets. Extended abstracts are not included in this volume.

The call for papers yielded 164 submissions, which were reviewed double-blind. Submitted work spanned from questions in statistical learning theory for geometric models, to novel geometric deep learning architectures, to methods for neural data analysis. Each submission was assigned a minimum of three reviews, which were aggregated and assessed by the editors for final inclusion. This resulted in 121 accepted works: 40 full-length papers, and 81 extended abstracts. All of the 40 full-length papers are published in this volume, by choice of the authors. Reviews of the accepted works can be found on the [NeurReps OpenReview Portal](#).

2.2. The Schedule

The workshop was held on December 7th, 2025 in San Diego, California, USA. The workshop featured six invited keynote speakers, with talks spanning topics in neuroscience, machine learning, topology, and their intersection, together with a discussion panel. All accepted works were presented as posters during the workshop, with a selection of submissions chosen for oral presentation. Best paper awards were given to the top contribution in each of three areas: *Graph, Geometric & Topological Deep Learning, Symmetry & Equivariance*, and *Neuroscience & Interpretability*. All talks and the panel were recorded and can be found online on the [NeurIPS virtual site](#).

KEYNOTES

Neural Oscillations in Machine Learning

Yue Song (Tsinghua University)

How Symmetry & Geometry Help Neural Representations Generalize

Max Tegmark (MIT)

Computation-Aware Representation Geometry Beyond Second-Order Statistics via Statistical Physics

SueYeon Chung (Harvard & Flatiron Institute)

New Mathematical Approaches to Interpretability and Robustness that Directly Confront the High Dimensionality and Nonlinearity of Neural Representations

Surya Ganguli (Stanford University)

When can ML models OOD generalize?

Razvan Pascanu (Google DeepMind & Mila)

Interpreting the neural code of the visual system using deep learning

Katrin Franke (Enigma Project)

DISCUSSION PANEL

Panel Discussion

Panelists: Surya Ganguli, T. Anderson Keller, Razvan Pascanu, Max Tegmark, SueYeon Chung, David Klindt

Moderator: Christian Shewmake



Figure 2: The NeurReps 2025 discussion panel.

CONTRIBUTED TALKS

From Finite to Infinite Groups: A Polynomial-Time Algorithm for Learning with Exact Invariances

Behrooz Tahmasebi

Any-Subgroup Equivariant Networks via Symmetry Breaking

Abhinav Goel

Beyond Pixels: A Differentiable Pipeline for Probing Neuronal Selectivity in 3D

Pavithra Elumalai

How does training shape the Riemannian geometry of neural network representations?

Jacob Zavatone-Veth

Causal Geometry of Batch Size and Generalisation

Zhongtian Sun

Koopman Autoencoders Learn Neural Representation Dynamics

Nishant Suresh Aswani

Measuring and Controlling Solution Degeneracy across Task-Trained Recurrent Neural Networks

Ann Huang

Balancing Fairness and Accuracy in Graph Learning via Fairness-Constrained Rewiring

Jason Wang

Emergent Riemannian geometry over learning discrete computations on continuous manifolds

Arthur Pellegrino

3. The Community

To support the growth of this nascent research area outside of the annual workshop, we have established a digital community for NeurReps, which at the time of writing has over 1,200 members. Instructions for joining and contributing to the community can be found on the [community page](#) of our website.

4. Program Committee

We extend our immense gratitude to our area chairs, who have diligently supervised the reviewing process and led the selection of contributed talks. Our heartfelt thanks also go out to the reviewers of our program committee, whose collective efforts resulted in the high-quality reviews for the workshop’s submissions. Their dedication and expertise have been integral to the success of this workshop. Thank you :

Area Chairs**Neuroscience
& Interpretability**

Meenakshi Khosla

**Topological
& Geometric ML**

Mathilde Papillon

**Symmetry
& Equivariance**

Soledad Villar

Reviewers

Abdelghani Azri	Daniel Platt	Leander Girrba
Adele Myers	Danil Akhtiamov	Louisa Cornelis
Aishwarya Balwani	David Robert Reich	Maghesree Chakraborty
Akshit Acharya	Davis Brown	Manu Madhav
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Arthur Pellegrino	Haitz Sáez de Ocáriz Borde	Prasad Kothari
Aslan Satary Dizaji	Hamed Karimi	Qingsong Wang
Bariscan Kurtkaya	Hamidreza Jamalabadi	Rahul Madhavan
Behrooz Tahmasebi	Hao Fang	Rana Muhammad Shahroz Khan
Berfin İnal	Hiba Sheheitli	Ratna Kandala
Bilal Alsallakh	Igor Caetano Diniz	Renjie Cao
Bo Zhao	Jacob A Zavattone-Veth	Rishabh Anand
Christos Zangos	Jeffrey Seely	Rishi Sonthalia
Colleen M. Farrelly	Jens Agerberg	S. Alireza Seyyed Mousavi
Congyue Deng	Jinen Setpal	Samy Blusseau
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Sina Tootoonian	Tomáš Karella	Yara Mansour
Soham Mukherjee	Torben Berndt	Yivan Zhang
Sonky Ung	Umang Garg	Yufan Wei
Sourabh Palande	Valentino Maiorca	Yuliang Guo
Srikanth Ramaswamy	Vasco Portilheiro	Zach Cohen
Subhadra Mokashe	Wei Ye	Zachary Lubberts
Sungjun Cho	Wenhao Zhang	Zakhar Shumaylov
Sungwoo Park	Wolfgang Polonik	

5. Moving Forward

We believe it is both timely and important to create a research venue and supportive community for the exchange of knowledge at the intersection of differential geometry, topology, machine learning and neuroscience. Moving forward, we will continue to create opportunities for dialogue and discussion on these themes at NeurIPS and other meetings. Furthering our broader aim of community-building, we have also established an active community of students and researchers which we believe will act as a gathering place to organize related events, such as seminars and hackathons.

References

- Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.
- Rishidev Chaudhuri, Berk Gerçek, Biraj Pandey, Adrien Peyrache, and Ila Fiete. The intrinsic attractor manifold and population dynamics of a canonical cognitive circuit across waking and sleep. *Nature neuroscience*, 22(9):1512–1520, 2019.
- Juan A Gallego, Matthew G Perich, Lee E Miller, and Sara A Solla. Neural manifolds for the control of movement. *Neuron*, 94(5):978–984, 2017.
- Richard J Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A Baas, Benjamin A Dunn, May-Britt Moser, and Edvard I Moser. Toroidal topology of population activity in grid cells. *Nature*, pages 1–6, 2022.
- Sung Soo Kim, Hervé Rouault, Shaul Druckmann, and Vivek Jayaraman. Ring attractor dynamics in the drosophila central brain. *Science*, 356(6340):849–853, 2017.
- Tanya Wolff, Nirmala A Iyer, and Gerald M Rubin. Neuroarchitecture and neuroanatomy of the drosophila central complex: A gal4-based dissection of protocerebral bridge neurons and circuits. *Journal of Comparative Neurology*, 523(7):997–1037, 2015.